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## PERSONAL SUMMARY

I am an environmental scientist with interdisciplinary training in geomorphology, hydrology, and physics. I have produced a variety of original research into geophysical transport phenomena, including river turbulence, granular flows, and sediment transport. My work includes both experiments and modeling, and it relies on techniques from many disciplines, including fluid dynamics, statistical mechanics, dynamical systems, image analysis, and computer vision. My ambition is to refine the interdisciplinary approach I have developed by contributing to a wide range of research problems that blur the boundaries between geoscience and physics.

## EDUCATION

**Ph.D. Geography**, The University of British Columbia (2021)

Specializations: Geomorphology and Hydrology

Dissertation: "The stochastic movements of individual streambed grains" (link).

Committee: Marwan Hassan (Supervisor), Rui Ferreira, Brett Eaton.

**M.Sc. Physics**, The University of British Columbia (2016)

Specialization: Condensed Matter Theory

Thesis: "Magnetic structure of chiral graphene nanoribbons" (link).

Supervisor: Ian Affleck

**B.S. Physics**, West Virginia University (2013)

GPA: 3.92/4.00 (Summa Cum Laude)

Award: "Outstanding Physics Senior" (2012)

Thesis: "A quantum field theory primer"

Minor: Mathematics (Analysis, Topology, Algebra)

## PUBLICATIONS

Submitted

1. **Pierce, K.** (2022). An advection-diffusion process with proportional resetting. [Submitted] Physical Review E, (preprint).
2. **Pierce, K.**, Hassan, M.A., & R.M.L. Ferreira (2022). Mechanistic description of the bed load sediment flux. [In Revision] Earth Surface Dynamics, (preprint).

Published

3. **Pierce, K.** & M.A. Hassan (2020). Back to Einstein: Burial-induced three-range diffusion in fluvial sediment transport. Geophysical Research Letters, 47 (15), 1-10, doi: 10.1029/2020GL087440, (paper).
4. **Pierce, K.** & M.A. Hassan (2020). Joint stochastic bedload transport and bed elevation model: Variance regulation and power-law rests. Journal of Geophysical Research: Earth Surface, 125 (4), 1–15. doi: 10.1029/2019JF005259, (paper).
5. Rowley J.D., **Pierce K.**, Brant A.T., Halliburton L.E., Giles N.C., Schunemann P.G., & A.D. Bristow (2012). Broadband terahertz pulse emission from $ZnGeP_2$. Optics Letters, 37 (5), 788-790. doi: 10.1364/OL.37.000788, (paper).

## RESEARCH EXPERIENCE

**Postdoctoral Research Fellow**, The University of British Columbia (Sep. 2021 – )

- Constructed laboratory flume experiments and developed mathematical models to understand the fundamental linkages between flow turbulence, granular collisions, and sediment transport at the grain scale. This work builds understanding of the fluid and granular physics underlying river stability and natural disaster risk.
- Prototyped many different experimental configurations to achieve simultaneous resolution of flow velocity fields and grain trajectories.
- Analyzed grain movements and fluid interfaces in 3D using computer vision techniques in Python, Matlab, and C++, including camera calibration, contrast equalization, perspective transformation, background subtraction, feature detection, object tracking, image matching, and contour finding.
- Resolved flow velocity fields using Acoustic Doppler and Particle Tracking Velocimetry. Calculated mean velocities, turbulent intensities, fluctuation spectra, and Reynolds stress profiles.

- Developed data analysis routines in Python, Labview, Matlab, and Bash which are used by other researchers in the lab. These routines clean and organize data, submit jobs to computing resources, acquire timed images, automate linear stages, measure bed topography with structured laser scanning, and determine sediment fluxes from visual occlusion.
- Automated data acquisition by coordinating linear stages, camera triggering, and flow velocity measurements in Labview, Matlab, and Python.
- Produced original stochastic models for geoscience, including the first description for the distribution of timescales over which sediment grains can remain stable in a turbulent flow, using a representation of sediment entrainment as a first passage problem.
- Presented findings at one conference and two invited talks. Submitted one manuscript to a leading physics journal, and several manuscripts are currently in preparation. Experimental results will be presented at two leading geophysics conferences this year.
- Secured funding for laboratory equipment by co-authoring, co-designing, and allocating the budget for an $187,488 CAD grant.

**Doctoral Research**, Mountain Channel Hydraulic Experimental Lab (2016 – 2021)

- Established new quantitative descriptions for the fluid-driven transport of coarse sediment The University of British Columbia. This work fuses probability and stochastic process theory, granular physics, and fluid dynamics ideas to provide fundamental understanding of sediment transport phenomena in river channels.
- Developed the first mathematical description for the multi-range anomalous diffusion displayed by transported sediment grains, based on a multi-state random walk to account for alternation of particles between motion, rest, and burial.
- Extended earlier birth-death-immigration-emigration type models of sediment transport at the grain scale to account for bed topography changes and their feedbacks with local sediment transport rates, based on a multi-species population model with non-linear competition.
- Provided the first description capable of predicting the full range of possible velocity distributions of particles propelled in a turbulent stream through repeated impacts with a granular bed, based on a stochastic representation of a randomly-driven granular gas.
- Formulated the first stochastic procedure to upscale from the Lagrangian trajectories of individual transported grains to a continuum-like sediment flux. This approach accounts for several poorly understood "scale-dependent" characteristics of sediment fluxes in river channels, based on the incorporation of renewal theory with earlier sediment trajectory models.
- Produced generalized random walk-type models to include entrainment and de-

position events into the dynamics of transported sediment particles for the first time, using stochastic processes with multiple noise sources.

- Implemented video game contact detection algorithms to conduct discrete element simulations of the dynamics of non-spherical grains in transport within a fluid flow.
- Presented research with 3 talks and 3 posters at the top geoscience conferences in North America and Europe.
- Published two research articles in leading geophysics journals: Geophysical Research Letters and Journal of Geophysical Research: Earth Surface. Another manuscript is currently in revision in a leading journal. Two manuscripts are in preparation, including an invited article for a special collection in Frontiers in Earth Science. Most of this research is summarized in the PhD dissertation.

**Master Research**, Advanced Materials and Process Engineering Lab (2014 – 2016)

- Evaluated the possibility of localized magnetism on the edges of thin ribbons of graphene in Ian Affleck's group at The University of British Columbia. The study was inspired by the possibility that such microscopic magnetism could be manipulated into quantum information storage in future computational devices.
- Achieved a proof of edge ferromagnetism on carbon ribbons with perfect edges, using analytical solutions of the Schrodinger equation in a tight-binding approximation.
- Extended the analytical results using numerical calculations performed in Mathematica and Python, showing that graphene ribbons with imperfect edges will also support edge magnetism.
- Presented these findings in two conference presentations and my master thesis. The theoretical conclusions in the work have since been experimentally confirmed.

**Undergraduate Research**, Ultrafast Nanophotonics Lab (2011 – 2012)

- Performed original research in non-linear optics and laser spectroscopy in the lab of Alan Bristow at WVU, examining (1) the relaxation dynamics of excited charge carriers in semiconductor crystals, and (2) the production of high-frequency (THz) light from optical rectification in chalcopyrite crystals.
- Constructed an optical experiment, built laser interferometers, tuned Ti-Sapphire pulsed lasers, and built a lock-in amplifier from scratch, starting from circuit diagrams and electronics components.
- Automated linear actuators and synced actuator movements to camera acquisition of laser signals using Labview.
- Secured a competitive $5000 USD National Science Foundation Award to fund

my summer work in the laboratory. The experiment I built this summer was later used by the lab to research photovoltaics for power generation.

- Presented findings on semi-conductor charge relaxation in a conference and co-authored a paper in Optics Letters reporting efficient optical rectification in a particular Zinc-Germanium-Phosphide crystal.

## PRESENTATIONS

Invited Talks

1. **Pierce, K.**, Hassan, M.A., & R.M.L. Ferreira (2022). A statistical mechanics description of rarefied sediment transport. University of Pennsylvania, Department of Earth and Environmental Sciences.

2. **Pierce, K.**, Moragoda, N., & L. Roberge (2021). Including wildfires in a landscape evolution model. Community Surface Dynamics Modeling System, Fall Webinar Series.

3. **Pierce, K.** & M.A. Hassan (2021). Velocity distributions, particle activities, and the sediment flux. Vanderbilt University, Department of Earth and Environmental Sciences.

4. **Pierce, K.** & M.A. Hassan (2020). Collisional Langevin approach to bed load sediment transport. Simon Fraser University, School of Environmental Science & Geography.

Conference Talks

5. **Pierce, K.** & M.A. Hassan (2021). Mechanistic-stochastic description of the bed-load sediment flux, AGU General Assembly Conference Abstracts, EP43A-02.

6. **Pierce, K.** (2021). Climate disasters and uncertain futures: Quantifying change in the environmental sciences, UBC Postdoctoral Research Day.

7. **Pierce, K.** & M.A. Hassan (2020). Collisional Langevin approach to bed load sediment velocity distributions, EGU General Assembly Conference Abstracts, EGU21-8148.

8. **Pierce, K.** & M.A. Hassan (2019). Back to Einstein: How to include trapping processes in fluvial diffusion models?, AGU Fall Meeting Abstracts, EP51B-02.

Conference Posters

9. **Pierce, K.** & M.A. Hassan (2020). Particle shape dictates critical shear stresses for sediment motion, AGU Fall Meeting Abstracts, EP013-0007.
10. Ferreira, R.M.L., Aleixo, R.F., Ricardo, A.M., **Pierce, K.**, & M.A. Hassan (2020) Turbulence in open-channel flows over mobile beds of high hydraulic conductivity, AGU Fall Meeting Abstracts, EP003-0002.
11. **Pierce, K.**, Ferreira, R.M.L., & M.A. Hassan (2019). Three-dimensional resolution of bedload transport with binocular computer vision, AGU Fall Meeting Abstracts, EP51F-2185.
12. **Pierce, K.** & I. Affleck (2015). Edge magnetism of chiral graphene nanoribbons, CIFAR Conference in Quantum Materials.
13. **Pierce, K.** & I. Affleck (2015). A rigorous proof of edge magnetism in graphene ribbons, Workshop on Frontiers of Condensed Matter Physics.
14. **Pierce, K.** & A. Bristow (2011). Hot-carrier dynamics in GaAs, WVNano Summer Undergraduate Research Symposium.

## GRANTS AND AWARDS

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Grants

1. **John R. Evans Leader Fund, Canada Foundation for Innovation** (2021 – )
   This grant secured funding for a new laboratory facility in the Mountain Channel Experimental Laboratory. I co-authored the research proposal, designed one component (among three) of the proposed facility, and prepared the itemized budget for that component (37% of the total budget). My work on grain-scale sediment transport provided preliminary data and design guidelines for the proposed facility. This grant funds my current laboratory work.
   - Project: Fluvial Landscapes Platform (LegoFlume), JERF 41979
   - Principal Investigator: Marwan Hassan
   - Role: Co-author
   - Amount: $187,488 CAD including matching funds

Funded Awards

2. **Earth Surface Processes Summer Institute Travel Fund** (2021)
   Competitive National Science Foundation award for travel to a Boulder, Colorado summer school in computational Earth Science. Amount: $800 USD.

3. **Outstanding Geography Teaching Assistant** (2020)
   Award provided to two teaching assistants annually (among approximately 80) in recognition of "significant contributions to undergraduate education". Amount: $500 CAD.

4. **Outstanding Physics Senior** (2013)
   Award provided to an "outstanding undergraduate majoring in physics who will graduate this year" upon faculty nomination. Amount: $500 USD.

5. **WVNano Summer Undergraduate Research Experience** (2012)
   Competitive National Science Foundation award for undergraduate research in quantum materials. Amount: $5000 USD.

6. **Edmund and Rose Rotter Endowed Scholarship** (2012)
   Department award provided to one undergraduate annually on the basis of "exceptional academic performance as determined by the Faculty of Science". Amount: $1500 USD.

## TEACHING EXPERIENCE

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**Research Mentor**, The University of British Columbia (2020 – )

- Supervised research activities of three students: (a) one undergraduate in Summer 2019, (b) one master student in Fall 2021, and (c) one PhD student from Fall 2021 to now.
- Guided students in laboratory work and statistical analysis tasks which culminated in respectively (a) a poster presentation of laboratory experiments, (b) a final course project based on field data, and (c) preliminary conclusions for a experimental fluid dynamics publication.

**Course Content Developer**, UBC Geography Department (2020-2021)

- Produced a 45 minute geomorphology field trip video for use in a 4th year undergraduate fluvial geomorphology course, in collaboration with a professor and audio visual technician. This film educates students on processes and features of mountain river channels.
- Developed five laboratory assignments and a virtual field trip, hosted in Tapestry, for a first year undergraduate geoscience course. Students quantitatively analyze mass wasting, volcanic processes, river morphology, climate change, and glacial processes.
- Produced video explanations of ecology and climate ideas to a virtual field trip phone application for a first year course on climate and ecosystems in collaboration with a professor. Students travel to public parks and learn from the video

explanations when they reach original filming locations. This effort was completed with grant funding obtained to incorporate emerging technologies into Earth Science education.

- Constructed five laboratory assignments for students in a third year "Statistics in Geography" course. Students analyze real-world pollution, ecology, climate, and hydrology problems during the assignments to learn collaborative R programming, version control, cloud computing, bash scripting, and inferential statistics methods.

**Geography Teaching Assistant**, The University of British Columbia (2016 – 2021)

- Produced assignments and exams, evaluated student work, hosted exams, conducted laboratory tutorials, performed classroom demonstrations, facilitated paper discussions, hosted office hours, delivered guest lectures, arranged student working groups, and led field trips. Managed educational technologies (Tapestry, Canvas, Blackboard, Github, Syzygy, Zoom, Slack), and evaluated student grades for 20 courses spanning all ranks of undergraduate and graduate study, totaling more than 3600 hours of contracted work.
- Mentored students on final course projects for 3rd year statistics (twice), 4th year hydrology and geomorphology, and graduate Fluvial Geomorphology courses. Guided topic selection, suggested research methodologies, moderated group work challenges, evaluated presentations, and monitored student progress toward project completion (Nearly 120 students total).
- Planned and delivered the entire tutorial component of a 16 week graduate course on Fluvial Geomorphology with 25 students. Presented lectures, designed original laboratory assignments, guided paper discussions, and evaluated student performance.
- Completed professional development courses including information security, inclusive teaching, classroom climate, lesson planning, time management, and workplace violence prevention.
- Received the UBC Geography "Outstanding TA" award in 2020 in recognition of "significant contributions to undergraduate education" in the UBC Geography department.

**Academic Coach**, UBC Varsity Athletics (2015 – 2017)

- Conducted one-on-one and group tutorial lessons on calculus, chemistry, physics, and statistics to reorient at-risk student athletes toward academic success.
- Mentored students in problem solving, effective communication, and academic goal setting.
- Prepared and distributed problem sets and example problems which enhanced student exam performance.

- Delivered academic skills workshops to groups of students on topics ranging from presentation skills to time management.

**Physics Teaching Assistant**, The University of British Columbia (2014 – 2015)

- Conducted laboratory and tutorial classes, evaluated student assignments and exams, hosted office hours, provided guest lectures, and administered exams in Classical Mechanics, Electromagnetism, Statistical Physics, and Quantum Mechanics courses spanning all undergraduate levels.

**Physics Teaching Assistant**, West Virginia University (2012 – 2013)

- Evaluated student assignments and exams, hosted office hours, and provided feedback on student performance for 2nd and 3rd year Mathematical Physics and Quantum Mechanics courses while a 4th year undergraduate.

## WORKSHOPS ATTENDED

**Earth Surface Processes Institute**, The University of Colorado, Boulder (2021)

- Attended an eight day workshop on Earth surface dynamics modeling.
- Studied numerical methods, best programming practices, open source software development, collaborative coding, version control, Landlab, high performance computing, and model uncertainty quantification.
- Developed a new computational landscape evolution model which evaluates impacts of wildfires on channel organization and sediment yields at a watershed scale in collaboration with two others.
- Contributed a Landlab-based "Education and Knowledge Transfer" lab, now hosted publicly on the Community Surface Dynamics Modeling System website (link) as a tutorial for the modeling community.
- Presented our work as a team at the "CSDMS Fall Webinar" online conference (link).

**Instructional Skills Workshop**, The University of British Columbia (2020)

- Participated in a carefully structured week long workshop on teaching models, strategies to foster active learning, inclusive teaching practices, interactive lesson planning, and emerging technologies in education.
- Delivered three lessons to a group of peers. Peer reviews and discussions of my lessons highlighted my own tendencies as an educator, helped accentuate my strong points, and helped me refine my weak points.

**Frontiers of Condensed Matter Physics**, The University of British Columbia (2015)

- Attended a two week workshop on condensed matter physics including dozens of lectures on topological materials, superconductivity, and frustrated magnetism.
- Collaborated on condensed matter physics problem sets, participated in discussions around open problems, and attended lectures.
- Presented my own research on low-dimensional magnetism at the workshop's conclusion.

## ACADEMIC SERVICE

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### Professional Activity

- **Manuscript Reviewing**: Geophysical Research Letters, Earth Surface Dynamics, Journal of Geophysical Research: Earth Surface, Water Resources Research, Advances in Water Resources.
- **Award Judging**: Outstanding Student Presentation Awards, American Geophysical Union Annual Meeting (2022).
- **Conference Chairing**: Session Chair, American Geophysical Union Annual Meeting (2022).
- **Seminar Organizing**: American Geophysical Union "Earth & Planetary Surface Processes Connects" seminar series (link) on a monthly basis (2021 – ).

### Institutional Activity

- **Workshop Presenting**: Produced workshops for graduate students in the department on scientific figure production in Inkscape and Matplotlib and manuscript preparation with LaTeX (2022).
- **Event Organizing**: Scheduled monthly department events over several years, including luncheons, "Earth-Surf Beers", and skill-share meetings (2020 – ).
- **Seminar Organizing**: Coordinated research group seminars for two years: secured dozens of visiting speakers, ordered catering, introduced speakers, led question and answer sessions, produced announcement fliers, sent announcement emails, and coordinated room and speaker scheduling. Established several joint seminars between The University of British Columbia and Simon Fraser University (2019 – 2021).

## COMMUNITY OUTREACH

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**Peer Support Volunteer**, Eastside Works Tech Cafe (2022 – )

- Volunteered in a community-led program to encourage tech literacy in Vancouver's under-served Downtown Eastside.
- Aided community members with technology challenges on a weekly basis. Challenges have included password management, resume writing, computer hardware problems, cell phone memory management, email usage, online banking tasks, tax filing, photo digitizing, and vaccine record locating.

**Volunteer Judge**, Vancouver Regional Science Fair (2021 – 2022)

- Evaluated science project presentations on a volunteer basis for middle and high school students, with intent to encourage their ongoing engagement in science.

**Tutorial Leader**, Science 101 (2021 – 2022)

- Volunteered in a university outreach program designed to introduce adults from traditionally-excluded communities to scientific thinking and academic opportunities.
- Led discussions and reflections on topics in Earth and Biological sciences, scientific writing, and critical reading. Delivered lectures on tectonics, identifying reliable sources, and plant taxonomy.

**Program Volunteer**, Books for Me! (2016 – 2018)

- Volunteered at a non-profit charity whose directive is to encourage literacy among disadvantaged children in Vancouver's downtown Eastside.
- Secured book donations, sorted books, and hosted "free book" distribution events once per week at different elementary schools during school terms.
- Aided children in selecting books to take home with intent to encourage childhood literacy.